

Concept 2 | Polynomial Division

English Student Edition • Focused formulas and guided practice

Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Worked Solutions

Polynomial Division

Q001 Challenge

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$. Then verify the result using $A = B \times Q + R$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 7x + 3 = (x^2 + x - 2) \times (2x - 1) + (-2x + 1)$.
4. Therefore, quotient $Q = 2x - 1$ and remainder $R = -2x + 1$.

Correct option: B

Final answer

$$Q = 2x - 1, \quad R = -2x + 1$$

Worked Solutions

Polynomial Division

Q002 Challenge

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x + 3) \times (3x - 1) + (x + 2)$.
3. Expand and collect like powers: $A = 3x^3 - 7x^2 + 12x - 1$.
4. Therefore, $A = 3x^3 - 7x^2 + 12x - 1$.

Correct option: A

Final answer

$$3x^3 - 7x^2 + 12x - 1$$

Worked Solutions

Polynomial Division

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3+5x^2+7x-7-(2x-4)}{x+3}$.
3. Divide and simplify: $B = x^2 + 2x - 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 + 5x^2 + 7x - 7$.

Correct option: D

Final answer

$$x^2 + 2x - 1$$

Worked Solutions

Polynomial Division

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

Solution

1. Use height = volume $\div$ base area: $h = \frac{2x^3+9x^2+4x-15}{x^2+2x-3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 5$.

Correct option: C

Final answer

$$2x + 5$$

Worked Solutions

Polynomial Division

Q005 Challenge

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - x^3 + 2x^2 - 4x + 1 = (x^2 - 2x + 2) \times (3x^2 + 5x + 6) + (-2x - 11)$.
4. Therefore, quotient $Q = 3x^2 + 5x + 6$ and remainder $R = -2x - 11$.

Correct option: B

Final answer

$$Q = 3x^2 + 5x + 6, \quad R = -2x - 11$$

Worked Solutions

Polynomial Division

Q006 Challenge

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^2 + 2x + 6 = (x + 2) \times (x^3 - 2x^2 + x) + (6)$.
4. Therefore, quotient $Q = x^3 - 2x^2 + x$ and remainder $R = 6$.

Correct option: A

Final answer

$$Q = x^3 - 2x^2 + x, \quad R = 6$$

Worked Solutions

Polynomial Division

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 - 5x + 1 = (x - 2) \times (x - 3) + (-5)$.
4. Therefore, quotient $Q = x - 3$ and remainder $R = -5$.

Final answer

$$Q = x - 3, \quad R = -5$$

Q008 Written

Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 4x + 1 = (x - 3) \times (x^2 + 3x + 5) + (16)$.
4. Therefore, quotient $Q = x^2 + 3x + 5$ and remainder $R = 16$.

Final answer

$$Q = x^2 + 3x + 5, \quad R = 16$$

Worked Solutions

Polynomial Division

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 2x^2 - 7x + 5 = (x - 2) \times (3x^2 + 8x + 9) + (23)$.
4. Therefore, quotient $Q = 3x^2 + 8x + 9$ and remainder $R = 23$.

Correct option: D**Final answer**

$$Q = 3x^2 + 8x + 9, \quad R = 23$$

Worked Solutions

Polynomial Division

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 6x^2 - 4 = (2x - 3) \times \left(\frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8} \right) + \left(\frac{103}{8} \right)$.
4. Therefore, quotient $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}$ and remainder $R = \frac{103}{8}$.

Correct option: B**Final answer**

$$Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, \quad R = \frac{103}{8}$$

Worked Solutions

Polynomial Division

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 4x^2 + 2x + 7 = (x - 1) \times (3x^2 - x + 1) + (8)$.
4. Therefore, quotient $Q = 3x^2 - x + 1$ and remainder $R = 8$.

Correct option: B**Final answer**

$$Q = 3x^2 - x + 1, \quad R = 8$$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 + x^3 - 5x + 4 = (x + 2) \times (2x^3 - 3x^2 + 6x - 17) + (38)$.
4. Therefore, quotient $Q = 2x^3 - 3x^2 + 6x - 17$ and remainder $R = 38$.

Correct option: A**Final answer**

$$Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$$

Worked Solutions

Polynomial Division

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^4 - 3x^3 + x - 6 = (x - 2) \times (4x^3 + 5x^2 + 10x + 21) + (36)$.
4. Therefore, quotient $Q = 4x^3 + 5x^2 + 10x + 21$ and remainder $R = 36$.

Correct option: D**Final answer**

$$Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$$

Worked Solutions

Polynomial Division

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 9x + 1 = (3x - 2) \times \left(\frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27} \right) + \left(-\frac{119}{27} \right)$.
4. Therefore, quotient $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}$ and remainder $R = -\frac{119}{27}$.

Correct option: B**Final answer**

$$Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$$

Worked Solutions

Polynomial Division

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 5x^2 - 2 = (2x^2 - x - 1) \times (x - 2) + (-x - 4)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = -x - 4$.

Final answer

$$Q = x - 2, \quad R = -x - 4$$

Q016 Written

Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 13x + 1 = (x^2 + 2x - 3) \times (2x - 4) + (x - 11)$.
4. Therefore, quotient $Q = 2x - 4$ and remainder $R = x - 11$.

Final answer

$$Q = 2x - 4, \quad R = x - 11$$

Worked Solutions

Polynomial Division

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 3x^2 + 3x + 4 = (x^2 - 2x + 3) \times (2x + 1) + (-x + 1)$.
4. Therefore, quotient $Q = 2x + 1$ and remainder $R = -x + 1$.

Final answer

$$Q = 2x + 1, \quad R = -x + 1$$

Q018 Written

Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 7x + 6 = (x^2 + 2x - 3) \times (x - 2) + (0)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = 0$.

Final answer

$$Q = x - 2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q019 MCQ

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 2x^2 - 5x + 4 = (x^2 + x - 2) \times (x + 1) + (-4x + 6)$.
4. Therefore, quotient $Q = x + 1$ and remainder $R = -4x + 6$.

Correct option: D**Final answer**

$$Q = x + 1, \quad R = -4x + 6$$

Q020 MCQ

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^3 + 2x - 5 = (x^2 + 2x - 1) \times (x^2 - 5x + 11) + (-25x + 6)$.
4. Therefore, quotient $Q = x^2 - 5x + 11$ and remainder $R = -25x + 6$.

Correct option: B**Final answer**

$$Q = x^2 - 5x + 11, \quad R = -25x + 6$$

Worked Solutions

Polynomial Division

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 4x + 5 = (x^2 - 2x + 2) \times (2x + 5) + (2x - 5)$.
4. Therefore, quotient $Q = 2x + 5$ and remainder $R = 2x - 5$.

Correct option: D**Final answer**

$$Q = 2x + 5, \quad R = 2x - 5$$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 2x^3 + 5x^2 - x + 3 = (x^2 + 3x - 1) \times (x^2 - 5x + 21) + (-69x + 24)$.
4. Therefore, quotient $Q = x^2 - 5x + 21$ and remainder $R = -69x + 24$.

Correct option: C**Final answer**

$$Q = x^2 - 5x + 21, \quad R = -69x + 24$$

Worked Solutions

Polynomial Division

Q023 MCQ

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 5x + 2 = (x^2 - x - 1) \times (3x + 3) + (x + 5)$.
4. Therefore, quotient $Q = 3x + 3$ and remainder $R = x + 5$.

Correct option: B**Final answer**

$$Q = 3x + 3, \quad R = x + 5$$

Q024 MCQ

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 - x^3 + 4x^2 - 7 = (x^2 + x + 3) \times (2x^2 - 3x + 1) + (8x - 10)$.
4. Therefore, quotient $Q = 2x^2 - 3x + 1$ and remainder $R = 8x - 10$.

Correct option: A**Final answer**

$$Q = 2x^2 - 3x + 1, \quad R = 8x - 10$$

Worked Solutions

Polynomial Division

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 8x^2 + 5 = (2x^2 - 2x - 5) \times (3x - 1) + (13x)$.
4. Therefore, quotient $Q = 3x - 1$ and remainder $R = 13x$.

Final answer

$$Q = 3x - 1, \quad R = 13x$$

Q026 Written

Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - 5x^3 + x - 1 = (-x^2 + 3x + 2) \times (-3x^2 - 4x - 18) + (63x + 35)$.
4. Therefore, quotient $Q = -3x^2 - 4x - 18$ and remainder $R = 63x + 35$.

Final answer

$$Q = -3x^2 - 4x - 18, \quad R = 63x + 35$$

Worked Solutions

Polynomial Division

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^3 - x^2 + 8x - 3 = (2x^2 + x - 4) \times \left(2x - \frac{3}{2}\right) + \left(\frac{35x}{2} - 9\right)$.
4. Therefore, quotient $Q = 2x - \frac{3}{2}$ and remainder $R = \frac{35x}{2} - 9$.

Correct option: D**Final answer**

$$Q = 2x - \frac{3}{2}, \quad R = \frac{35x}{2} - 9$$

Q028 MCQ

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 8x + 2 = (x^2 - 4) \times (x) + (-4x + 2)$.
4. Therefore, quotient $Q = x$ and remainder $R = -4x + 2$.

Correct option: B**Final answer**

$$Q = x, \quad R = -4x + 2$$

Worked Solutions

Polynomial Division

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $5x^3 - 4x^2 + x - 6 = (2x^2 - x + 1) \times \left(\frac{5x}{2} - \frac{3}{4}\right) + \left(-\frac{9x}{4} - \frac{21}{4}\right)$.
4. Therefore, quotient $Q = \frac{5x}{2} - \frac{3}{4}$ and remainder $R = -\frac{9x}{4} - \frac{21}{4}$.

Correct option: D**Final answer**

$$Q = \frac{5x}{2} - \frac{3}{4}, \quad R = -\frac{9x}{4} - \frac{21}{4}$$

Worked Solutions

Polynomial Division

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 + 2x^3 - x^2 + 5 = (4x^2 + x - 2) \times \left(\frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64} \right) + \left(\frac{37x}{64} + \frac{163}{32} \right)$.
4. Therefore, quotient $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}$ and remainder $R = \frac{37x}{64} + \frac{163}{32}$.

Correct option: C**Final answer**

$$Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, \quad R = \frac{37x}{64} + \frac{163}{32}$$

Worked Solutions

Polynomial Division

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 4x^2 - x + 9 = (2x^2 - x + 3) \times \left(\frac{3x}{2} + \frac{11}{4}\right) + \left(-\frac{11x}{4} + \frac{3}{4}\right)$.
4. Therefore, quotient $Q = \frac{3x}{2} + \frac{11}{4}$ and remainder $R = -\frac{11x}{4} + \frac{3}{4}$.

Correct option: D**Final answer**

$$Q = \frac{3x}{2} + \frac{11}{4}, \quad R = -\frac{11x}{4} + \frac{3}{4}$$

Worked Solutions

Polynomial Division

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 5x + 1 = (2x^2 + 3x - 2) \times \left(3x - \frac{9}{2}\right) + \left(\frac{29x}{2} - 8\right)$.
4. Therefore, quotient $Q = 3x - \frac{9}{2}$ and remainder $R = \frac{29x}{2} - 8$.

Correct option: B

Final answer

$$Q = 3x - \frac{9}{2}, \quad R = \frac{29x}{2} - 8$$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-9a^2 - 9ax + 4x^2 = (-3a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q034 Written

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-5a^3 + 5a^2x - 3ax^2 + x^3 = (-2a + x) \times (3a^2 - ax + x^2) + (a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = a^3$$

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-6a^2 - 5ax + 4x^2 = (-2a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q036 Written

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-a^3 + a^2x + ax^2 + x^3 = (2a + x) \times (3a^2 - ax + x^2) + (-7a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = -7a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = -7a^3$$

Q037 Written

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3a^3 - 2ax^2 + x^3 = (a + x) \times (3a^2 - 3ax + x^2) + (0)$.
4. Therefore, quotient $Q = 3a^2 - 3ax + x^2$ and remainder $R = 0$.

Final answer

$$Q = 3a^2 - 3ax + x^2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-2a^3 + 7a^2x - 4ax^2 + x^3 = (-a + x) \times (4a^2 - 3ax + x^2) + (2a^3)$.
4. Therefore, quotient $Q = 4a^2 - 3ax + x^2$ and remainder $R = 2a^3$.

Correct option: D

Final answer

$$Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$$

Q039 MCQ

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 + 3xy - 4y^2 = (x + y) \times (x + 2y) + (-6y^2)$.
4. Therefore, quotient $Q = x + 2y$ and remainder $R = -6y^2$.

Correct option: B

Final answer

$$Q = x + 2y, \quad R = -6y^2$$

Worked Solutions

Polynomial Division

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 3x^2y - 2y^3 = (x + y) \times (x^2 + 2xy - 2y^2) + (0)$.
4. Therefore, quotient $Q = x^2 + 2xy - 2y^2$ and remainder $R = 0$.

Final answer

$$Q = x^2 + 2xy - 2y^2, \quad R = 0$$

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (x + 3) + (-2x + 5)$.
3. Expand and collect like powers: $A = x^3 + x^2 - 9x + 2$.
4. Therefore, $A = x^3 + x^2 - 9x + 2$.

Final answer

$$x^3 + x^2 - 9x + 2$$

Worked Solutions

Polynomial Division

Q042 Written

Find the polynomial **A** when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (2x - 3) + (-2x)$.
3. Expand and collect like powers: $A = 2x^3 - 7x^2 + 2x + 3$.
4. Therefore, $A = 2x^3 - 7x^2 + 2x + 3$.

Final answer

$$2x^3 - 7x^2 + 2x + 3$$

Q043 Written

Find the polynomial **A** when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 3x + 2) \times (2x + 6) + (2x - 3)$.
3. Expand and collect like powers: $A = 2x^3 - 12x + 9$.
4. Therefore, $A = 2x^3 - 12x + 9$.

Final answer

$$2x^3 - 12x + 9$$

Worked Solutions

Polynomial Division

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 - x + 2) \times (2x - 1) + (2)$.
3. Expand and collect like powers: $A = 4x^3 - 4x^2 + 5x$.
4. Therefore, $A = 4x^3 - 4x^2 + 5x$.

Final answer

$$4x^3 - 4x^2 + 5x$$

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 + x + 1) \times (3x + 2) + (-2x + 6)$.
3. Expand and collect like powers: $A = 6x^3 + 7x^2 + 3x + 8$.
4. Therefore, $A = 6x^3 + 7x^2 + 3x + 8$.

Final answer

$$6x^3 + 7x^2 + 3x + 8$$

Worked Solutions

Polynomial Division

Q046 Written

When $4x^3 + 8x - 9$ is divided by B , the quotient is $2x - 3$ and the remainder is $x + 15$. Find B .

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{4x^3+8x-9-(x+15)}{2x-3}$.
3. Divide and simplify: $B = 2x^2 + 3x + 8$.
4. Check by multiplying B by Q and adding R to recover $A = 4x^3 + 8x - 9$.

Final answer

$$2x^2 + 3x + 8$$

Q047 Written

When $x^3 - x^2 + 3x + 1$ is divided by B , the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B .

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3-x^2+3x+1-(-4)}{x^2-2x+5}$.
3. Divide and simplify: $B = x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 - x^2 + 3x + 1$.

Final answer

$$x + 1$$

Worked Solutions

Polynomial Division

Q048 Written

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+x^2-3x+5-(-5x+2)}{x+1}$.
3. Divide and simplify: $B = 2x^2 - x + 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + x^2 - 3x + 5$.

Final answer

$$2x^2 - x + 3$$

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B, the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{6x^3+11x^2-2-(-1)}{2x^2+3x-1}$.
3. Divide and simplify: $B = 3x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = 6x^3 + 11x^2 - 2$.

Final answer

$$3x + 1$$

Worked Solutions

Polynomial Division

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B, the quotient is $x + 2$ and the remainder is $3x + 13$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+4x^2+7-(3x+13)}{x+2}$.
3. Divide and simplify: $B = 2x^2 - 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + 4x^2 + 7$.

Final answer

$$2x^2 - 3$$

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B, the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^4+x^2+3x-1-(3x-2)}{x^2+x+1}$.
3. Divide and simplify: $B = x^2 - x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^4 + x^2 + 3x - 1$.

Final answer

$$x^2 - x + 1$$

Worked Solutions

Polynomial Division

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

Solution

1. Use height = volume $\div$ base area: $h = \frac{4x^3 + 4x^2 - 9x - 9}{2x^2 - x - 3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 3$.

Final answer

$$2x + 3$$

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R ?:

Solution

1. Long division gives $R = -5$.

Correct option: A

Final answer

$$-5$$

Worked Solutions

Polynomial Division

Quiz MCQ 2 **Concept Quiz · MCQ**

In a polynomial in x , a is treated as... Which of the following is correct?:

Solution

1. Only powers of x determine the division order.

Correct option: B

Final answer

constant

Quiz MCQ 3 **Concept Quiz · MCQ**

To recover B from $A = B \times Q + R$, use... Which of the following is correct?:

Solution

1. Isolate B after moving R to the other side.

Correct option: C

Final answer

$$\frac{A - R}{Q}$$

Worked Solutions

Polynomial Division

Quiz MCQ 4 Concept Quiz · MCQ

If B has degree 2, a valid remainder can have degree...:

Solution

1. The remainder degree must be below 2.

Correct option: C

Final answer

1

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q, R, and verify $A = B \times Q + R$.

Solution

1. Long division gives $Q = 3x^2 + x + 4$, $R = 1$.
2. Substitution verifies $3x^3 - 5x^2 + 2x - 7 = (x - 2) \times (3x^2 + x + 4) + (1)$.

Final answer

$$Q = 3x^2 + x + 4, \quad R = 1$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 6 Concept Quiz · Written

Recover A when $B=x^2 - x - 2$, $Q=2x + 3$, $R=-x + 1$.

Solution

1. Use $A = B \times Q + R$.
2. $A = 2x^3 + x^2 - 8x - 5$.

Final answer

$$2x^3 + x^2 - 8x - 5$$

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Solution

1. Divide volume by base area: $h = 2x + 3$.
2. The remainder is zero, so the height is polynomial.

Final answer

$$2x + 3$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 8 Concept Quiz · Written

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Solution

1. Long division gives $Q = x^2 - x - 2$ and $R = 2x + 1$.
2. The remainder degree is less than 2, so it is valid even though it is nonzero.

Final answer

$$Q = x^2 - x - 2, \quad R = 2x + 1$$

Answer key

Use the question number shown on each page.

Q001 B	Q016 $Q = 2x - 4$, $R = x - 11$	Q031 D	Q046 $2x^2 + 3x + 8$
Q002 A	Q017 $Q = 2x + 1$, $R = -x + 1$	Q032 B	Q047 $x + 1$
Q003 D	Q018 $Q = x - 2$, $R = 0$	Q033 $Q = 3a + 4x$, $R = 0$	Q048 $2x^2 - x + 3$
Q004 C	Q019 D	Q034 $Q = 3a^2 - ax + x^2$, $R = a^3$	Q049 $3x + 1$
Q005 B	Q020 B	Q035 $Q = 3a + 4x$, $R = 0$	Q050 $2x^2 - 3$
Q006 A	Q021 D	Q036 $Q = 3a^2 - ax + x^2$, $R = -7a^3$	Q051 $x^2 - x + 1$
Q007 $Q = x - 3$, $R = -5$	Q022 C	Q037 $Q = 3a^2 - 3ax + x^2$, $R = 0$	Q052 $2x + 3$
Q008 $Q = x^2 + 3x + 5$, $R = 16$	Q023 B	Q038 D	Quiz MCQ 1 A
Q009 D	Q024 A	Q039 B	Quiz MCQ 2 B
Q010 B	Q025 $Q = 3x - 1$, $R = 13x$	Q040 $Q = x^2 + 2xy - 2y^2$, $R = 0$	Quiz MCQ 3 C
Q011 B	Q026 $Q = -3x^2 - 4x - 18$, $R = 63x + 35$	Q041 $x^3 + x^2 - 9x + 2$	Quiz MCQ 4 C
Q012 A	Q027 D	Q042 $2x^3 - 7x^2 + 2x + 3$	Quiz WRITTEN 5 $Q = 3x^2 + x + 4$, $R = 1$
Q013 D	Q028 B	Q043 $2x^3 - 12x + 9$	Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Q014 B	Q029 D	Q044 $4x^3 - 4x^2 + 5x$	Quiz WRITTEN 7 $2x + 3$
Q015 $Q = x - 2$, $R = -x - 4$	Q030 C	Q045 $6x^3 + 7x^2 + 3x + 8$	Quiz WRITTEN 8 $Q = x^2 - x - 2$, $R = 2x + 1$